Problem Set 11 – Two-Way Analysis of Variance

Problem 1 (15 points)
Please access the *Retention* data. These data and the context are borrowed from Purdue University. Here are the details, quoted from the Purdue source:

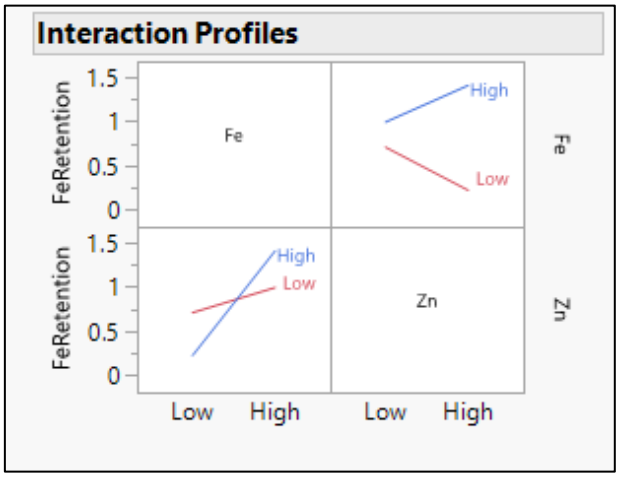
> Iron and Zinc fortification of milk-based fruit drinks are common practice. To better understand the effects of drink fortification on the cellular retentionof iron, researchers conducted an experiment by fortifying milk-based fruit drinks with low and high levels of iron (Fe) and zinc (Zn). The drinks were digested in a simulated gastroinatestinal tract and cellular iron retention was measured (mg Fe/mg cell protein).

Please answer each of the following question. So that your answer remains focused, you are to include ONLY the item(s) requested (else, your earned credit will be reduced).

(15 points) A colleague in your department performed a two-way ANOVA on these data and concluded that "both Iron and Zinc tested as important" for cellular iron retention. They have asked if you agree. The results of their JMP output are displayed below. Following that output, and using 100 words or less, give a well-thought out answer to your colleague about the importance of these two factors.

Effect Tests

Source	Nparm	DF	Sum of Squares	F Ratio	Prob > F
Fe	1	1	4.4022863	2355.702	<.0001*
Zn	1	1	0.0108781	5.8210	0.0226*
Fe*Zn	1	1	1.6554721	885.8576	<.0001*



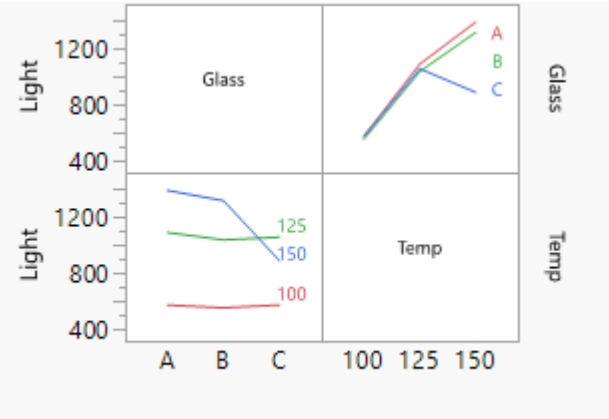
I agree that both iron and zinc significantly affect cellular iron retention. Iron ($p < 0.0001$) and zinc ($p = 0.0226$) are both statistically significant at the 0.05 level. However, the interaction between iron and zinc is also significant ($p < 0.0001$), indicating that the effect of one factor depends on the level of the other. The interaction plot supports this because the lines are not parallel. Therefore, both factors are important, but their interaction must also be considered when interpreting the results.

Problem 2 (35 points)
Please access the *GTL* data which present the results of an experiment conducted to study the influence of the operating temperature (100˚C, 125˚C and 150˚C) and three faceplate glass types (A, B and C) in the light output of an oscilloscope tube. They are from Design and Analysis of Experiments, 8th edition, by Douglas Montgomery.

Please answer each of the following questions. So that your answers remain focused, you are to include ONLY the item(s) requested (else, your earned credit will be reduced).

a. (5 points) Fit a two-way ANOVA model with light output (continuous) as the response and temperature (character) and glass type (character) as factors. Include an interaction term. Copy and paste the Effect Tests table and the Interaction Profiles plot below.

Source	Nparm	DF	Sum of Squares	F Ratio	Prob > F
Glass	2	2	150864.5	206.3706	<.0001
Temp	2	2	1970334.5	2695.259	<.0001
Glass*Temp	4	4	290551.7	198.7257	<.0001



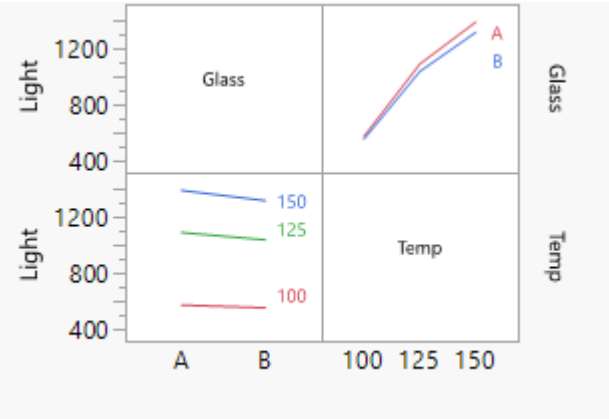
b. (5 points) Refer to part a. In 100 words or less, explain whether the effect of 'operating temperature' is dependent on 'glass type'.

Yes, the effect of operating temperature on light output depends on glass type. The interaction between glass type and temperature is statistically significant ($p < 0.0001$), indicating that different glass types respond differently to changes in temperature. The interaction plot shows that light output increases with temperature for glass types A and B,

while glass type C decreases in light output between 125°C and 150°C. Therefore, the effect of temperature is dependent on the type of glass being used.

c. (5 points) Fit a two-way ANOVA model with light output (continuous) as the response, and temperature (character) and glass type (character) as factors. However, ONLY include glass types A and B. Do NOT include type C. Include an interaction term in the model. Copy and paste the Effect Tests table and the Interaction Profiles plot below.

Source	Nparm	DF	Sum of Squares	F Ratio	Prob > F
Glass	1	1	10512.5	29.6591	0.0001
Temp	2	2	1900633.3	2681.144	<.0001
Glass*Temp	2	2	2169.3	3.0602	0.0843



d. (5 points) Refer to part c. Just by looking at the Interaction Profiles plot (a plot of the treatments means), using 100 words or less, defend the Glass-Temperature combination that would likely produce maximum light.

Based on the Interaction Profiles plot, Glass Type A at 150°C would likely produce the maximum light output. The plot shows that light output increases as temperature increases for both glass types. However, Glass Type A produces slightly more light than Glass Type B at 150°C. Therefore, Glass Type A at 150°C would be the best combination for maximizing light output.

e. (5 points) Refer to part c. Take an alpha level of 0.05. In 100 words or less, explain why you should drop the interaction term and rerun the model.

The interaction between Glass Type and Temperature is not statistically significant because its p-value (0.0843) is greater than the significance level of 0.05. This indicates there is insufficient evidence that the effect of temperature on light output depends on glass type. Therefore, the interaction term should be removed, and the model should be rerun using only the main effects of Glass Type and Temperature.

f. (10 points) Refer to part d. Rerun the model (no type C glass) without the interaction

term. In 100 words or less, explain whether Glass Type or Temperature Level, or both, are important.

Both Glass Type and Temperature are statistically significant factors affecting light output. Temperature ($p < 0.0001$) and Glass Type ($p = 0.0003$) are both significant at the 0.05 level. Temperature has a much larger F Ratio (2071.481) than Glass Type (22.9149), suggesting that temperature has a greater influence on light output. Therefore, both factors are important, but temperature appears to have the stronger effect.